SQL – Stock Maintenance Query Questions

1. SELECT & WHERE

1. Display all products from the stock table

```
1  18:46:16  CREATE DATABASE stock_db      1 row(s) affected   0.016 sec
2  18:46:25  USE stock_db                  0 row(s) affected   0.000 sec
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	quantity	price	supplier	reorder_level
1	Laptop	Electronics	8	55000.00	ABC_Suppliers	10
2	Monitor	Electronics	25	8000.00	ABC_Suppliers	10
3	Keyboard	Electronics	15	1500.00	ABC_Suppliers	10
4	Chair	Furniture	50	2500.00	Papu_Furniture	20
5	Table	Furniture	5	7000.00	Papu_Furniture	10
6	Notebook	Stationery	100	100.00	D_Mart	30
7	Diary	Stationery	200	200.00	D_Mart	50
8	Speaker	Electronics	6	12000.00	ABC_Suppliers	5
9	Telivision	Electronics	7	40000.00	ABC_Suppliers	15
10	Sofa	Furniture	20	10000.00	Papu_Furniture	10
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Result Grid | Form Editor | Field Types | stock 4 | Apply | Revert

2. Display the product name, category, and price of all products

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

product_name	category	price
Laptop	Electronics	55000.00
Monitor	Electronics	8000.00
Keyboard	Electronics	1500.00
Chair	Furniture	2500.00
Table	Furniture	7000.00
Notebook	Stationery	100.00
Diary	Stationery	200.00
Speaker	Electronics	12000.00
Telivision	Electronics	40000.00
Sofa	Furniture	10000.00

Result Grid | Form Editor | Field Types

3. Display products whose quantity is less than 10.

```
27 •  SELECT * FROM stock WHERE quantity<10;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	quantity	price	supplier	reorder_level
1	Laptop	Electronics	8	55000.00	ABC_Suppliers	10
5	Table	Furniture	5	7000.00	Papu_Furniture	10
8	Speaker	Electronics	6	12000.00	ABC_Suppliers	5
9	Telivision	Electronics	7	40000.00	ABC_Suppliers	15
NULL	NULL	NULL	NULL	NULL	NULL	NULL

stock 6 | Apply

4. Display products whose price is greater than 1000.

```
28 •  SELECT * FROM stock WHERE price>1000;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	quantity	price	supplier	reorder_level
1	Laptop	Electronics	8	55000.00	ABC_Suppliers	10
2	Monitor	Electronics	25	8000.00	ABC_Suppliers	10
3	Keyboard	Electronics	15	1500.00	ABC_Suppliers	10
4	Chair	Furniture	50	2500.00	Papu_Furniture	20
5	Table	Furniture	5	7000.00	Papu_Furniture	10
8	Speaker	Electronics	6	12000.00	ABC_Suppliers	5
9	Telivision	Electronics	7	40000.00	ABC_Suppliers	15
10	Sofa	Furniture	20	10000.00	Papu_Furniture	10
NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. Display products belonging to the Electronics category.

```
29 •  SELECT * FROM stock WHERE category='Electronics';
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	quantity	price	supplier	reorder_level
1	Laptop	Electronics	8	55000.00	ABC_Suppliers	10
2	Monitor	Electronics	25	8000.00	ABC_Suppliers	10
3	Keyboard	Electronics	15	1500.00	ABC_Suppliers	10
8	Speaker	Electronics	6	12000.00	ABC_Suppliers	5
9	Telivision	Electronics	7	40000.00	ABC_Suppliers	15
NULL	NULL	NULL	NULL	NULL	NULL	NULL

stock 8 × Apply

2. ORDER BY

1. Display products in ascending order of price.

```
29 •  SELECT * FROM stock WHERE category= Electronics ;
30 •  SELECT * FROM stock ORDER BY price ASC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	quantity	price	supplier	reorder_level
6	Notebook	Stationery	100	100.00	D_Mart	30
7	Diary	Stationery	200	200.00	D_Mart	50
3	Keyboard	Electronics	15	1500.00	ABC_Suppliers	10
4	Chair	Furniture	50	2500.00	Papu_Furniture	20
5	Table	Furniture	5	7000.00	Papu_Furniture	10
2	Monitor	Electronics	25	8000.00	ABC_Suppliers	10
10	Sofa	Furniture	20	10000.00	Papu_Furniture	10
8	Speaker	Electronics	6	12000.00	ABC_Suppliers	5
9	Telivision	Electronics	7	40000.00	ABC_Suppliers	15
1	Laptop	Electronics	8	55000.00	ABC_Suppliers	10
NULL	NULL	NULL	NULL	NULL	NULL	NULL

2. Display products in descending order of price

```
30 •  SELECT * FROM stock ORDER BY price ASC;
31 •  SELECT * FROM stock ORDER BY price DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cel

product_id	product_name	category	quantity	price	supplier	reorder_level
1	Laptop	Electronics	8	55000.00	ABC_Suppliers	10
9	Telivision	Electronics	7	40000.00	ABC_Suppliers	15
8	Speaker	Electronics	6	12000.00	ABC_Suppliers	5
10	Sofa	Furniture	20	10000.00	Papu_Furniture	10
2	Monitor	Electronics	25	8000.00	ABC_Suppliers	10
5	Table	Furniture	5	7000.00	Papu_Furniture	10
4	Chair	Furniture	50	2500.00	Papu_Furniture	20
3	Keyboard	Electronics	15	1500.00	ABC_Suppliers	10
7	Diary	Stationery	200	200.00	D_Mart	50
6	Notebook	Stationery	100	100.00	D_Mart	30
NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Display products in ascending order of quantity.

```
32 •  SELECT * FROM stock ORDER BY quantity ASC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	quantity	price	supplier	reorder_level
5	Table	Furniture	5	7000.00	Papu_Furniture	10
8	Speaker	Electronics	6	12000.00	ABC_Suppliers	5
9	Telivision	Electronics	7	40000.00	ABC_Suppliers	15
1	Laptop	Electronics	8	55000.00	ABC_Suppliers	10
3	Keyboard	Electronics	15	1500.00	ABC_Suppliers	10
10	Sofa	Furniture	20	10000.00	Papu_Furniture	10
2	Monitor	Electronics	25	8000.00	ABC_Suppliers	10
4	Chair	Furniture	50	2500.00	Papu_Furniture	20
6	Notebook	Stationery	100	100.00	D_Mart	30
7	Diary	Stationery	200	200.00	D_Mart	50
NULL	NULL	NULL	NULL	NULL	NULL	NULL

Result Grid | Form Editor | Field Types

stock 11 | Apply | Revert

2. Display products in descending order of price.

```
33 •  SELECT * FROM stock ORDER BY quantity DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Co

product_id	product_name	category	quantity	price	supplier	reorder_level
7	Diary	Stationery	200	200.00	D_Mart	50
6	Notebook	Stationery	100	100.00	D_Mart	30
4	Chair	Furniture	50	2500.00	Papu_Furniture	20
2	Monitor	Electronics	25	8000.00	ABC_Suppliers	10
10	Sofa	Furniture	20	10000.00	Papu_Furniture	10
3	Keyboard	Electronics	15	1500.00	ABC_Suppliers	10
1	Laptop	Electronics	8	55000.00	ABC_Suppliers	10
9	Telivision	Electronics	7	40000.00	ABC_Suppliers	15
8	Speaker	Electronics	6	12000.00	ABC_Suppliers	5
5	Table	Furniture	5	7000.00	Papu_Furniture	10
NULL	NULL	NULL	NULL	NULL	NULL	NULL

4. Display products by category in ascending order and price in descending order.

```
33 •  SELECT * FROM stock ORDER BY category ASC,price DESC;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell

product_id	product_name	category	quantity	price	supplier	reorder_level
1	Laptop	Electronics	8	55000.00	ABC_Suppliers	10
9	Telivision	Electronics	7	40000.00	ABC_Suppliers	15
8	Speaker	Electronics	6	12000.00	ABC_Suppliers	5
2	Monitor	Electronics	25	8000.00	ABC_Suppliers	10
3	Keyboard	Electronics	15	1500.00	ABC_Suppliers	10
10	Sofa	Furniture	20	10000.00	Papu_Furniture	10
5	Table	Furniture	5	7000.00	Papu_Furniture	10
4	Chair	Furniture	50	2500.00	Papu_Furniture	20
7	Diary	Stationery	200	200.00	D_Mart	50
6	Notebook	Stationery	100	100.00	D_Mart	30
NULL	NULL	NULL	NULL	NULL	NULL	NULL

5. Display the 5 most expensive products.

```
34 •  SELECT * FROM stock ORDER BY price DESC
35    LIMIT 5;
```

Result Grid | Filter Rows: | Edit: | Export/Import: | Wrap Cell Content:

product_id	product_name	category	quantity	price	supplier	reorder_level
1	Laptop	Electronics	8	55000.00	ABC_Suppliers	10
9	Telivision	Electronics	7	40000.00	ABC_Suppliers	15
8	Speaker	Electronics	6	12000.00	ABC_Suppliers	5
10	Sofa	Furniture	20	10000.00	Papu_Furniture	10
2	Monitor	Electronics	25	8000.00	ABC_Suppliers	10
NULL	NULL	NULL	NULL	NULL	NULL	NULL

3. Aggregate Functions

1. Find the total number of products.

```
41 • SELECT category, COUNT(*)AS product_count FROM stock GROUP BY category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	product_count
Electronics	5
Furniture	3
Stationery	2

3. Find the average product price.

```
43 • SELECT category, AVG(price)AS average_price FROM stock GROUP BY category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	average_price
Electronics	23300.000000
Furniture	6500.000000
Stationery	150.000000

2. Find the total available quantity

```
42 • SELECT category, SUM(quantity)AS total_quantity FROM stock GROUP BY category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_quantity
Electronics	61
Furniture	75
Stationery	300

4. Find the highest product price.

```
44 •  SELECT category, MAX(price)AS highest_price FROM stock GROUP BY category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	highest_price
Electronics	55000.00
Furniture	10000.00
Stationery	200.00

4. GROUP BY

1. Find the number of products in each category.

```
41 •  SELECT category, COUNT(*)AS product_count FROM stock GROUP BY category;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	product_count
Electronics	5
Furniture	3
Stationery	1

5. Find the total number of products supplied by each supplier.

```
45 •  SELECT supplier, COUNT(*)AS product_count FROM stock GROUP BY supplier;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

supplier	product_count
ABC_Suppliers	5
Papu_Furniture	3
D_Mart	2

4. Find the highest-priced product in each category.

```
46 •  SELECT category, COUNT(*)AS more_products FROM stock GROUP BY category HAVING COUNT(*) >=5;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	more_products
Electronics	5

Result Grid | Form Editor

2. Find the total quantity available in each category.

```
47 •  category, SUM(quantity) AS total_quantity FROM stock GROUP BY category HAVING SUM(quantity) > 100;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	total_quantity
Stationery	300

Result Grid | Form Editor

3. Find the average price of products in each category.

```
48 •  SELECT category, AVG(price)AS average_price FROM stock GROUP BY category HAVING AVG(price)>1000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	average_price
Electronics	23300.000000
Furniture	6500.000000

Result Grid | Form Editor

4. Display suppliers supplying more than 3 products.

```
49 •  SELECT supplier, COUNT(*)AS more_products FROM stock GROUP BY supplier HAVING COUNT(*)>3;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

supplier	more_products
ABC_Suppliers	5

Result Grid | Form Editor

5. Display categories where the highest product price is greater than 5000.

```
50 •  SELECT category, MAX(price)AS highest_price FROM stock GROUP BY category HAVING MAX(price)>5000;
```

Result Grid | Filter Rows: | Export: | Wrap Cell Content:

category	highest_price
Electronics	55000.00
Furniture	10000.00

Result Grid | Form Editor | Field Types

2. Increase the price of all products by 10%.

```
51 •  UPDATE  stock SET price=100000 WHERE product_id=1;
```

Output

Action Output

#	Time	Action	Message
48	21:53:14	SELECT supplier, COUNT(*)AS more_products FROM stock GROUP BY supplier HAVING C...	1 row(s) returned
49	21:55:41	SELECT category, MAX(price)AS highest_price FROM stock GROUP BY category HAVING ...	2 row(s) returned
50	22:00:42	UPDATE stock SET price=100000 WHERE product_id=1	1 row(s) affected Rows matched: 1 Cha

4. Change the supplier of a particular product.

```
53 •  UPDATE stock SET supplier='samin' WHERE product_id=1;
```

Output

Action Output

#	Time	Action	Message
54	22:05:46	UPDATE stock SET price=price * 1.10	Error Code: 1175. You are using safe update
55	22:08:19	UPDATE stock SET supplier='samin' WHERE product_name='laptop'	Error Code: 1175. You are using safe update
56	22:09:01	UPDATE stock SET supplier='samin' WHERE product_id=1	1 row(s) affected Rows matched: 1 Change

1. Delete a product using its product_id.

```
54 •  DELETE FROM stock WHERE  product_id=7;
55 •  SELECT * FROM stock;
```

Result Grid | Filter Rows: | Edit: | Export/Import:

product_id	product_name	category	quantity	price	supplier	reorder_level
1	Laptop	Electronics	8	100000.00	samin	10
2	Monitor	Electronics	25	8000.00	ABC_Suppliers	10
3	Keyboard	Electronics	15	1500.00	ABC_Suppliers	10
4	Chair	Furniture	50	2500.00	Papu_Furniture	20
5	Table	Furniture	5	7000.00	Papu_Furniture	10
6	Notebook	Stationery	100	100.00	D_Mart	30
8	Speaker	Electronics	6	12000.00	ABC_Suppliers	5
9	Telivision	Electronics	7	40000.00	ABC_Suppliers	15
10	Sofa	Furniture	20	10000.00	Papu_Furniture	10
NULL	NULL	NULL	NULL	NULL	NULL	NULL